Gaps Between Public Transport Need and Accessibility in Urban Areas of Israel

Final Project — Geographic Information and Artificial Intelligence

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ArcGIS Online project:

https://infoi.maps.arcgis.com/apps/mapviewer/index.html?webmap=3f7b8a254d78433e8053a5c3381434a7

Google Colab code:

https://drive.google.com/file/d/1qendT_WFNsLzhzkZg-2t7rmnnRbtTMks/view?usp=drive_link

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1. Introduction

Public transport plays an important role in urban development, reducing social inequalities, and providing access to essential services such as employment, education, healthcare, and commerce. The level of public transport accessibility directly affects residents' quality of life and their ability to participate in economic and social activity, especially among groups that depend more heavily on public transport, including households without a private car, older adults, and lower-income populations.

Geographic Information Systems (GIS) make it possible to combine spatial, demographic, and transport data and identify areas where the needs of the population are not matched by the level of service available. By combining data from Israel's Central Bureau of Statistics (CBS) with national public transport data in GTFS format, it is possible to perform a spatial assessment of public transport accessibility and evaluate whether the existing service corresponds to local needs.

The purpose of this project is to identify urban statistical areas in Israel with the largest gap between the need for public transport and the level of accessibility provided. The analysis combines 2022 census data from the CBS with nationwide GTFS public transport data and applies spatial analysis using ArcGIS Pro. The results were also published in ArcGIS Online to provide an interactive visualization of the spatial patterns identified in the study.

2. Literature Review

2.1 Currie (2010)

Currie, G. (2010). Quantifying Spatial Gaps in Public Transport Supply Based on Social Needs. Journal of Transport Geography, 18(1), 31–41.

Problem and research objective

The study examines spatial gaps between population needs and public transport service levels in Melbourne, Australia. Currie notes that much previous work relied on qualitative information or resident reports, leaving a need for an objective and quantitative method for identifying areas where public transport supply is insufficient. The study therefore develops a GIS-based methodology for systematically measuring the gap between social need and public transport supply.

Model and methodology

The study developed two main indicators. The first was a need index, combining the official Australian IRSAD socioeconomic index with a Transport Needs Index based on demographic and social characteristics. The second was a supply index based on walking distance to stops, service frequency, and spatial coverage. GIS was used to create buffers around public transport stops, combine census and transport data, and compare the two indices in order to identify areas where high need coincides with inadequate service.

Main findings

The analysis identified census areas in Melbourne where very high public transport need was combined with insufficient service and estimated the population living in those areas. These gaps were especially common in outer suburban areas, although several inner-city areas also showed meaningful deficiencies. This indicates that accessibility problems are not limited to the metropolitan periphery.

Did the study meet its objective?

Yes. The paper presented a clear quantitative GIS-based method for measuring the gap between population needs and public transport supply and demonstrated how it can be used to identify areas that should be prioritized for service improvements and transport planning.

Personal assessment

In my view, the main strength of the study is its relatively clear and practical methodology, which can be adapted to other locations. At the same time, the model assumes that population is evenly distributed within each census area and does not account for residents' actual travel destinations. As a result, some differences may remain between the modelled accessibility patterns and real travel behavior.

Contribution to this project

Currie's study provided the main methodological basis for this project. In a similar way, the current analysis calculates separate need and supply indices and then measures the gap between them to identify areas with high need and low public transport accessibility. Both approaches also rely on GIS, buffers around stops, and the integration of census and public transport data.

2.2 Bocarejo & Oviedo (2012)

Bocarejo, J. P., & Oviedo, D. R. (2012). Transport accessibility and social inequities: A tool for identification of mobility needs and evaluation of transport investments. Journal of Transport Geography, 24, 142–154.

Problem and research objective

This paper examines public transport accessibility and social inequality in Bogotá, Colombia. The authors argue that transport systems should not be evaluated only through travel-time reductions or infrastructure expansion. Evaluation should also consider actual access to opportunities and the financial burden placed on users, particularly among disadvantaged populations. The study develops a tool for measuring accessibility and supporting the prioritization of transport investment and policy.

Model and methodology

The authors developed an accessibility model based on the Hansen equation, with an impedance function that incorporates both travel time and transport cost as a share of the user's income. They distinguish between Real Accessibility, Accessibility under Standard Parameters, and Available Accessibility under Desired Preferences. The analysis uses Bogotá's 2005 origin–destination survey, employment and income data, and GIS for spatial analysis and visualization.

Main findings

Residents of lower-income neighborhoods were found to spend more time travelling and a substantially larger share of their income on public transport, while still having lower access to employment opportunities. The study also found that in some cases, fare subsidies could improve accessibility more effectively than infrastructure expansion alone.

Did the study meet its objective?

Yes. The paper presents a quantitative accessibility framework that combines spatial and economic considerations and demonstrates how it can be used to evaluate transport projects and prioritize investments in a more equitable way.

Personal assessment

A major strength of the study is the integration of spatial accessibility with affordability, which provides a broader view of how transport affects different population groups. However, the model is based on data from a single city, so adjustments may be required before applying it to locations with different urban, social, or transport characteristics.

Contribution to this project

This study reinforced the importance of including social and economic factors when analyzing public transport accessibility. Like the Bogotá study, my project combines demographic and transport data to identify locations where population needs are not matched by service levels. The main difference is that this project calculates a Need–Supply Gap for urban statistical areas across Israel using CBS and GTFS data, while Bocarejo and Oviedo focus on accessibility to employment opportunities within Bogotá.

3. Data and Methods

The project combines data from three main sources: the 2022 CBS statistical-area layer, the 2022 population census, and nationwide GTFS public transport data from Israel's Ministry of Transport. The census variables used in the analysis include population size, the share of households without a car, the share of residents aged 65 and over, and median annual income.

The first stage consisted of data processing in Python using Google Colab, with AI assistance used for code writing and technical troubleshooting. Census data were joined to the statistical-area layer, urban residential areas were filtered, and GTFS data were processed to calculate the number of departures from each stop on a representative weekday. A 400-meter buffer was then created around public transport stops. For each statistical area, the analysis calculated stop coverage and departures per capita, and then constructed need, supply, and gap indices. A Getis-Ord Gi* hot-spot analysis was also performed, and three GeoJSON files were produced for further work in ArcGIS Pro.

In the second stage, the GeoJSON layers were imported into ArcGIS Pro for spatial analysis and cartographic design. Appropriate symbology, class breaks, transparency, layer order, display ranges, and legends were configured for each layer. ArcGIS Pro's Hot Spot Analysis (Getis-Ord Gi*) tool was also used as an additional validation step and to generate the final hot-spot layer. A Mean Gap by Locality layer was created as well, showing the average gap score for each locality using graduated point size and color.

Finally, four map layouts were produced: the Public-Transport Need Index, the Need–Supply Gap and Critical Areas, statistically significant gap clusters, and Mean Gap by Locality. The layers and maps were then published to ArcGIS Online so that the results could be explored interactively.

4. Findings

The analysis covered 2,179 urban statistical areas across 142 localities in Israel. CBS population data were combined with national public transport data to identify areas where the level of public transport need is not matched by accessibility and service supply.

The Public-Transport Need Index map (Figure 1, Appendix B) shows how need varies among statistical areas. The index is based on the share of households without a car, the proportion of residents aged 65 and over, population density, and median income. The results show substantial variation between areas, including differences within the same locality.

The Need–Supply Gap map (Figure 2, Appendix B) represents the difference between the need and supply indices. Physical stop coverage is high in most urban areas: median coverage is approximately 100% and average coverage is about 94%. This means that the main factor distinguishing areas is service frequency and the number of available departures, rather than simply whether a stop is located nearby.

The analysis identified 168 statistical areas classified as "high need – low accessibility." These areas provide the most direct answer to the research question because they combine relatively high dependence on public transport with a comparatively low level of service.

The hot-spot map (Figure 3, Appendix B) identifies statistically significant spatial concentrations of accessibility gaps. The strongest concentration appears in Tel Aviv–Yafo, where 121 of 161 statistical areas—about 75%—were classified as hot spots. Additional concentrations were found in Jerusalem and in other urban areas in northern and southern Israel. Cold spots, representing a relatively better match between need and supply, were particularly visible in Netanya, Bnei Brak, and several Arab localities, including Umm al-Fahm, Tayibe, Tamra, Sakhnin, and Tira.

The Mean Gap by Locality map (Figure 4, Appendix B) summarizes the results at the municipal level and allows comparison between local authorities. The map shows that meaningful gaps exist not only in peripheral regions but also in dense urban centers. The overall pattern therefore cannot be explained simply as a center-versus-periphery divide.

5. Discussion and Conclusions

The findings show that CBS data, GTFS data, and GIS tools can be combined to systematically identify areas where public transport need is not matched by accessibility. In most urban areas, physical access to stops is already high; the more important factor affecting actual supply is service frequency.

The results are consistent with the main ideas in the reviewed literature. As in Currie (2010), need and supply were calculated separately and then compared through a gap measure. In line with Bocarejo and Oviedo (2012), the analysis also emphasizes the importance of evaluating public transport from the perspective of populations with greater mobility needs rather than focusing only on infrastructure coverage.

The study has several limitations. It relies on scheduled GTFS data rather than observed service performance and assumes a uniform 400-meter walking distance to stops. In addition, the demographic indicators are based on 2022 data and therefore do not reflect later changes in either population characteristics or the public transport network.

Overall, the project answered the research question and successfully identified areas with substantial mismatches between population needs and public transport service. Combining Python-based data processing, spatial analysis in ArcGIS Pro, and publication through ArcGIS Online produced a clear set of maps and a practical analytical tool that can help identify areas where public transport service improvements should be considered.

Bibliography

Currie, G. (2010). Quantifying spatial gaps in public transport supply based on social needs. Journal of Transport Geography, 18(1), 31–41.

Bocarejo, J. P., & Oviedo, D. R. (2012). Transport accessibility and social inequities: A tool for identification of mobility needs and evaluation of transport investments. Journal of Transport Geography, 24, 142–154.

Israel Central Bureau of Statistics. (2022). Population Census and 2022 Statistical Areas Layer.

Israel Ministry of Transport and Road Safety. Public transport data in GTFS format.

Appendix A – AI Prompts Used During the Project

1. Formulating the research question

> *I want to conduct a project on public transport availability in Israel. Help me formulate a research question.*

2. Selecting data sources

> *What data do I need to examine the gap between the need for public transport and accessibility to it in Israel? Suggest relevant data sources from the CBS and the Ministry of Transport.*

3. Designing the analysis workflow

> *Suggest a Python workflow that combines CBS statistical-area and census data with GTFS data in order to calculate public transport need, supply, and gap indices.*

4. Creating GIS layers

> *Help me write Python code that calculates the indices and produces GeoJSON files that I can import into ArcGIS Pro.*

5. Working in ArcGIS Pro

> *I have GeoJSON layers containing need, supply, and gap indices. Suggest which analyses, layers, and symbology I should create in ArcGIS Pro to answer the research question.*

6. Selecting variables for the need index

> *Which variables from the population census should be included in a public transport need index, and how should they be combined?*

7. Hot-spot analysis

> *I want to identify spatial clusters of areas with a high gap between need and supply. Which GIS tool is suitable for this, and how should I use it?*

8. Checking result quality

> *Review the results of the need, supply, and gap indices and look for outliers or implausible values that might indicate a problem in the processing workflow.*

9. Solving the data-join problem

> *I cannot identify the correct join field between the population census data and the statistical-area layer. How can I determine the correct field from the data itself?*

Appendix B – Result Maps

The following four layouts were produced in ArcGIS Pro and were also published in ArcGIS Online.

Figure 1: Public-Transport Need Index

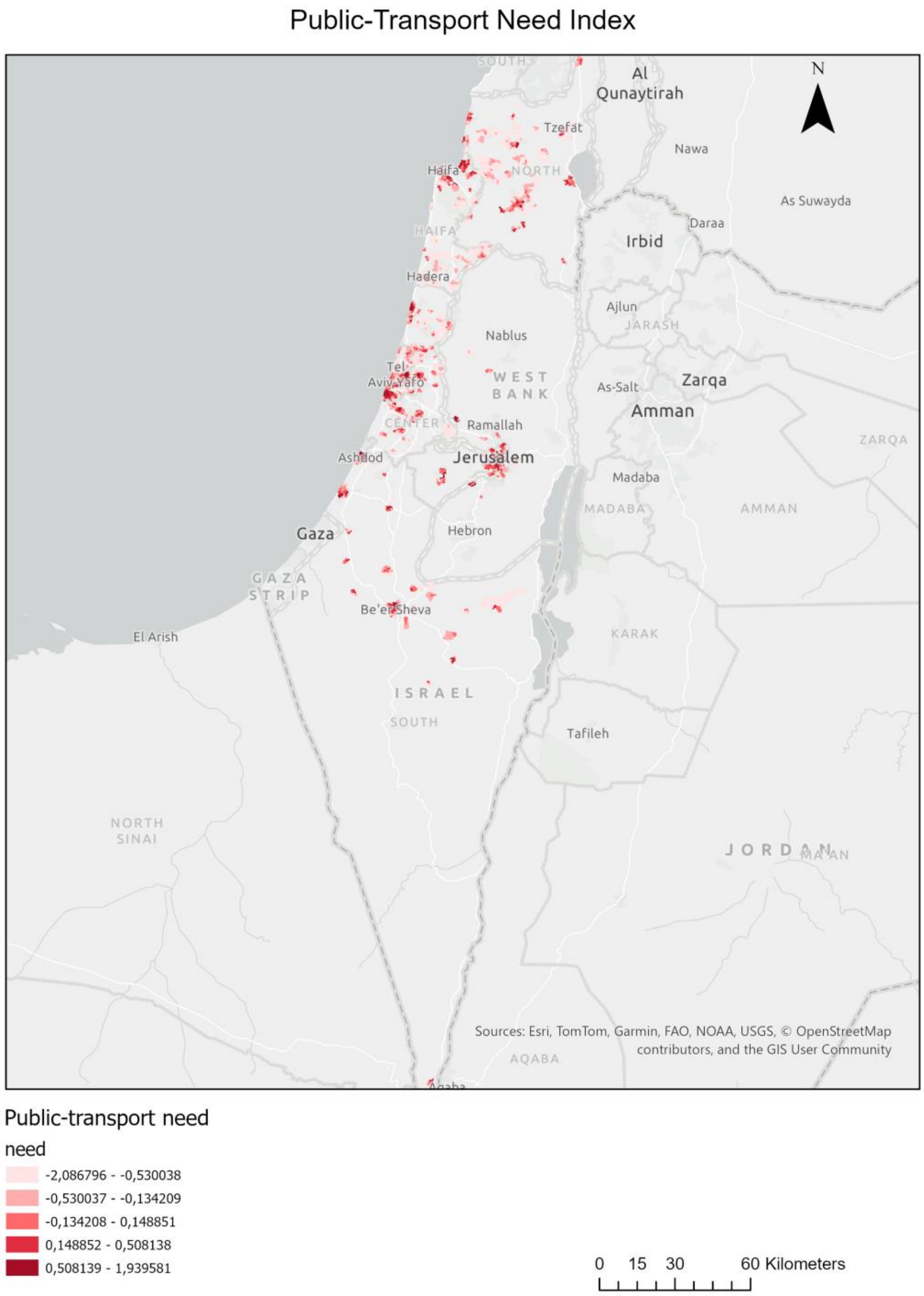

Figure 2: Need–Supply Gap and Critical Areas

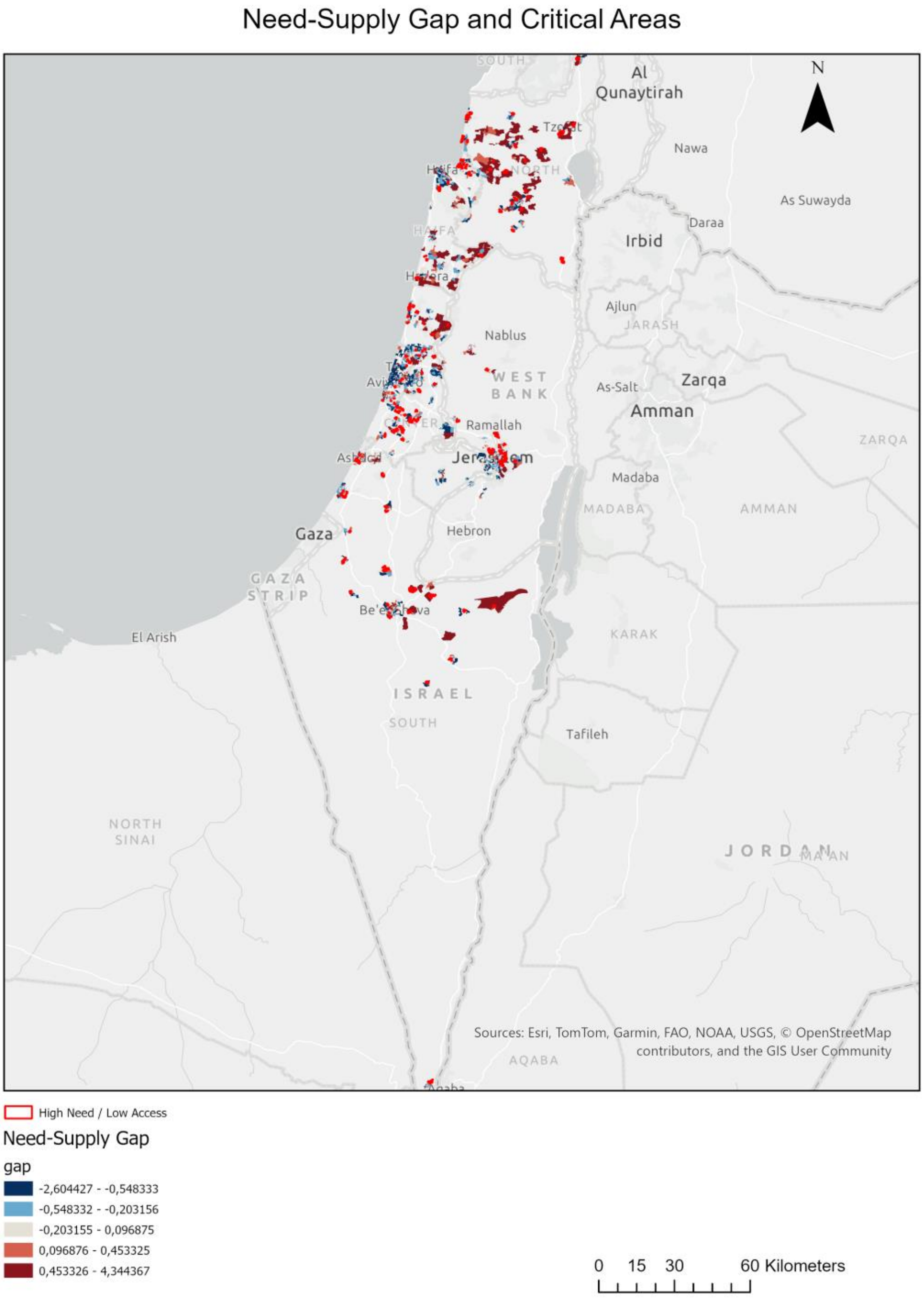

Figure 3: Statistically Significant Gap Clusters – Getis-Ord Gi*

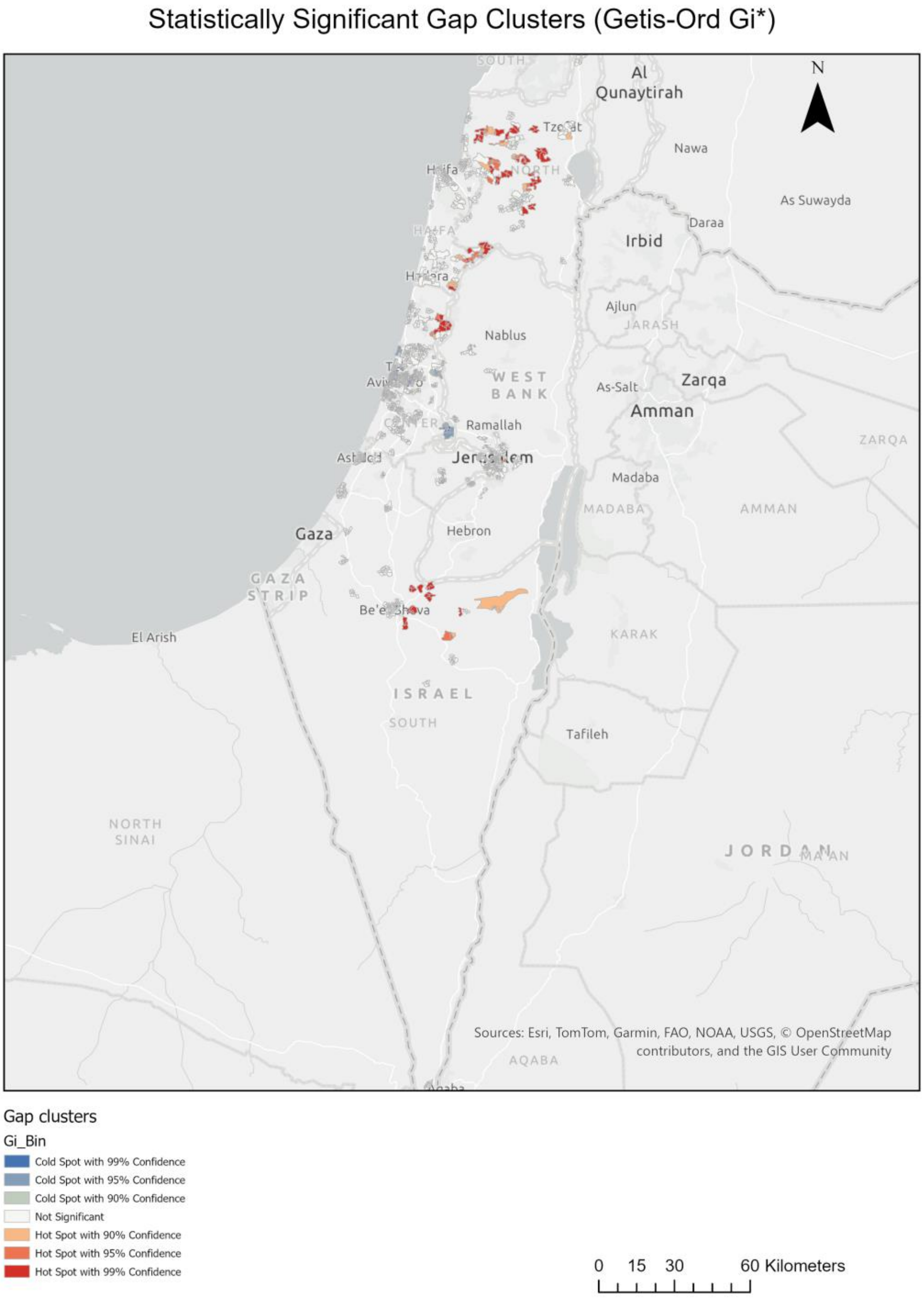

Figure 4: Mean Gap by Locality